FS Exam Prep — GUI Phase Spec (Tabs + De-brand) v1.0.0 — 2026-10-21

Purpose: Phase 2 of the app architecture Charlie approved 2026-10-20 (engine connection first — done, §7 audit 6/6 PASS on commit `48e7e50` — then GUI). This phase has two blocks, run in one AI Studio session, in order, with one GitHub push at the end:

- **Block A — Structure:** replace the single stacked home screen with a 4-tab bottom-navigation layout. Structure only: **no feature may be removed, disabled, or altered.**
- **Block B — De-brand + stale copy:** remove the name "NCEES" from all user-facing text (trademark/liability — Charlie's call, 2026-10-20), remove two false "cut score" claims, and fix two home-screen strings left stale by the engine integration.

Base state (verified in code, commit `48e7e50`): `HomeScreen.kt` is 1,057 lines stacking title, streak, daily-goal progress, badges preview, leaderboard preview, mock-exam card, adaptive card, all domain rows, completed modules, and the theme picker on one scroll. Navigation is a `ScreenState` state machine in the ViewModel (HOME, QUIZ, SCORE_SUMMARY, MOCK_EXAM, MOCK_EXAM_RESULT, ACHIEVEMENTS, LEADERBOARD, FLASHCARDS, STUDY_GUIDE) with **no bottom navigation bar**. The floating action button (FAB) on the home screen opens Quiz Configuration. Calm Pro theme tokens are in place and must not change.

---

1. Block A — Tab restructure

1.1 Navigation

- Add a bottom navigation bar with exactly four tabs: **Today, Practice, Learn, Progress**.
- The bar is visible on the four tab screens and **hidden** during quiz, mock exam, mock-exam result, and score summary flows.
- Every existing feature must remain reachable in ≤ **2 taps** from a tab screen.

1.2 Feature → tab mapping (complete inventory — nothing may be dropped)

The current home screen's full feature surface (from its callbacks and sections) maps as follows:

Feature (current)	Destination
Greeting / app title area	Today (title becomes "FS Exam Prep" per Block B)
Streak chip + overall progress % + Daily Goal Progress	Today — one readiness summary card
Primary action	Today — ONE button: "Continue studying" (resume most relevant next action) or "Start today's session"
Weakest domain ("Needs Work" from domain stats)	Today — one "Up next" row with a Start action into that domain's quiz
Domain quiz list — all 7 domains, with per-domain strength/accuracy	Practice

Feature (current)	Destination
Adaptive Practice ("Start Adaptive Practice (10 Qs)")	Practice
Mock Exam card + Start Mock Exam	Practice
Quiz Configuration (currently the FAB's destination)	Practice
Study Guide	Learn
Flashcards	Learn
Detailed stats (per-domain breakdown)	Progress
Badges / Achievements (preview count + full screen)	Progress
Leaderboard (preview + full screen)	Progress
Completed Practice Modules (review answers/explanations)	Progress
Appearance Theme picker	Settings gear icon in the top app bar, shown on all four tab screens
Non-affiliation disclaimer (new, Block B §2.3)	Progress — small footer line

1.3 Today tab — minimal, in this order, nothing else

1. Short greeting line with the user's handle.
2. One readiness summary card: overall progress percentage, current streak, daily-goal progress.
3. ONE primary button ("Continue studying" / "Start today's session").
4. One "Up next" row: weakest domain + Start action.

No domain lists, tool grids, achievement or leaderboard previews, or settings on Today.

1.4 Practice / Learn / Progress tabs

- Simple flat lists with dividers, one action per row, in the Calm Pro style. No new card stacks, gradients, or decorative elements.
- Progress rows lead to the existing full screens (Achievements, Leaderboard, module review) unchanged.

1.5 FAB

- Remove the floating action button. Its action (Quiz Configuration) lives in the Practice tab per §1.2.

1.6 Unchanged by Block A

- The quiz, mock exam, mock-exam result, flashcards, study guide, achievements, leaderboard, and score summary screens themselves — only how the user reaches them changes.
- All ViewModel behavior, scoring logic, and repository behavior.

---

2. Block B — De-brand "NCEES" + stale copy

"NCEES" is a third-party organization's trademark. The app must never appear affiliated with, endorsed by, or sourced from it. There is also **no published fixed cut score** for the FS exam — the app must not state or imply one.

2.1 Exact replacements (verified occurrences, commit `48e7e50`)

#	File	Current	Replacement
B1	HomeScreen.kt	`NCEES FS Exam Prep` (title)	`FS Exam Prep`
B2	HomeScreen.kt	`Full-Length NCEES FS Mock Exam`	`Full-Length FS Mock Exam`
B3	MockExamScreen.kt	`NCEES FS CBT Simulation`	`FS Exam Simulation`
B4	MockExamResultScreen.kt	`Score: X% • You exceeded the NCEES cut score threshold.`	`Score: X% • A passing-level performance.`
B5	MockExamResultScreen.kt	`Performance across all NCEES exam specifications`	`Performance across all FS exam domains`
B6	ScoreSummaryScreen.kt	`Well done! You surpassed the NCEES cut score threshold.`	`Well done! A passing-level performance.`
B7	QuizRepository.kt	Achievement: `Complete a full timed NCEES FS Mock Exam`	`Complete a full timed FS Mock Exam`
B8	GamificationModels.kt	Code comment `typically >= 70% on NCEES FS`	`typically >= 70% (practice benchmark)`
B9	metadata.json	`NCEES Fundamentals of Surveying (FS) practice questions, …`	`Fundamentals of Surveying (FS) practice questions, domain quizzes, strength & weakness analytics, and detailed problem solutions.`

2.2 Stale mock-card copy (left behind by the engine integration — same fix pass)

#	File	Current	Replacement
B10	HomeScreen.kt	`Timed 90m`	`Timed 5h 20m`
B11	HomeScreen.kt	`32 exam questions covering all 8 domains per official specifications. Includes live countdown timer, question palette, flagging, and domain diagnostic report.`	`110 exam questions covering all 7 FS domains. Includes live countdown timer, question palette, flagging, and domain diagnostic report.`

2.3 Disclaimer (new)

Add this exact line as a small footer on the Progress tab:

Independent study tool. Not affiliated with or endorsed by NCEES. All questions are original practice material.

2.4 Metadata honesty (same file as B9)

`metadata.json` also declares `MAJOR_CAPABILITY_SERVER_SIDE_GEMINI_API`, which the app does not use. Remove that capability entry.

2.5 Catch-all

If any other user-facing string contains "NCEES", apply the same pattern: describe the FS exam generically; never use the organization's name as a brand, a question source, or a score authority. Do not change any question content, scoring logic, or features in Block B.

---

3. Guardrails (both blocks — non-negotiable)

1. **Off-limits files:** `QuestionBank.kt` and everything under `data/` that loads the bank, the bank JSON under `assets/`, `FsDomain.kt`, and the theme files (`Color.kt`, `Type.kt`, `Theme.kt`). The engine connection verified in the §7 audit must not be disturbed.
2. **BankErrorScreen behavior stays:** if the bank fails validation, the error screen still replaces the app UI.
3. **Calm Pro tokens unchanged** — colors, typography (Noto Serif for Learn headings), flat divider-based style.
4. **No new dependencies** (no navigation library — implement tabs with the existing state machine / Compose primitives already in the project).
5. **No feature removed or behavior changed** — Block A moves things; Block B changes strings only (plus the §2.3 footer and §2.4 metadata entry).

---

4. Acceptance checks (Muse audits the push against these)

1. Bottom navigation exists with exactly Today / Practice / Learn / Progress; hidden during quiz, mock, result, and summary flows.
2. Feature inventory (§1.2) complete: every listed feature reachable in ≤ 2 taps; Today contains only the §1.3 elements.
3. FAB gone; Quiz Configuration reachable from Practice.
4. Theme picker no longer on a tab screen; settings gear in the top app bar on all four tab screens.
5. `NCEES` count in user-facing strings and code comments: **zero** (the §2.3 disclaimer's "endorsed by NCEES" is the single allowed occurrence — it is a non-affiliation statement).
6. Both cut-score claims replaced per B4/B6; no other string states or implies a fixed cut score.
7. Mock card reads "Timed 5h 20m" and the B11 description.
8. Disclaimer present verbatim (§2.3) on Progress.
9. metadata.json: B9 description applied; Gemini capability entry removed.
10. Guardrail diff check: QuestionBank/bank assets/FsDomain/theme files untouched; no new dependencies in build files.